CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

SENI NODE SCORING MODEL V2

Five-Dimension Prioritization Framework with Hard-Stop Conditions, Evidence-Tiered D2, and Weight Sensitivity Analysis

NOTICE *Independent policy research. All sources open-source. This document represents independent analytical assessment and does not constitute official policy direction or legal advice. Scoring model reflects the author's analytical judgment applied to open-source evidence; it does not represent an official prioritization by any government agency.*

KEY JUDGMENTS

KJ 1

Tier 1 nodes (Immediate Action): N43 NIIAR (ADJ:4.75), N33 NCCP (ADJ:4.05), N34 ChMP (ADJ:3.55). These 3 nodes satisfy all combined Tier 1 conditions: D1≥4, D2≥3, D3≥3, adj≥3.50, no hard-stop. Rankings are stable across all weight perturbations tested (±10% on D2).

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Hard-stop conditions correctly constrain UECC (N28): D3=1 triggers catastrophic collateral hard-stop, capping at Tier 2 regardless of composite score. This is the model working as designed — highest-leverage node cannot be Tier 1 until supply precondition is met.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 3

D2 evidence tiering revision: N33 and N43 revised from D2=4 (v1) to D2=3. No named correspondent banks for Isotope JSC are documented in open-source evidence. D2=3 reflects confirmed category-level Western exposure without named counterparties. This revision is a downgrade in evidentiary confidence, not a downgrade in the underlying financial leverage — the leverage exists, the documentation does not yet.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 4

10 nodes correctly land in Tier 3 including all weapons-complex nodes with closed defense budgets (D2=1 hard-stop), the Manzhouli crossing (D1=0 hard-stop), and the analytically inferred transport nodes (LOW-MOD confidence modifier). The model is structurally sound and defensible to hostile review.

ALMOST CERTAINLY — HIGH CONFIDENCE

I. VERSION 2 CHANGES — RESPONSE TO POST-REVIEW FIXES

FIX	CHANGE	RATIONALE
D2 Evidence Tiering + FPA Integration	D2 rubric distinguishes [5]=named banks+flows; [4]=named counterparties; [3]=confirmed category. N33 and N43 upgraded to D2=4 [4+] per FPA-01-v3: Wells Fargo Bank N.A. named as Lantheus banking institution via SEC S-1 (EDGAR CIK 0001521036); clearing pathway inferred not transaction-confirmed. D2=[4+] = upper boundary of OSINT-accessible financial attribution. ADJ scores: N43 4.50 (HIGH); N33 4.05 (MOD-HIGH, ×0.90). Both remain firmly Tier 1.	Correctly scopes OSINT ceiling: named bank confirmed; clearing link inferred. D2=[4+] notation preserves evidentiary discipline while acknowledging the near-complete attribution achieved. See FPA-01-v3.
Hard-Stop Conditions	Binary override conditions applied after confidence adjustment: D3≤1 → Tier 2 ceiling; D2≤1 → Tier 2 ceiling; D1=0 → Tier 3 ceiling; D4=0 → Tier 3 ceiling.	Resolves 'some dimensions are gating constraints, not additive variables' critique. UECC (D3=1) cannot reach Tier 1 regardless of other scores.
Dimension-Floor Tier Logic	Tier 1 requires D1≥4 AND D2≥3 AND D3≥3 AND adj≥3.50. Thresholds now tied to decision logic, not arbitrary numbers.	Resolves 'thresholds are arbitrary' critique. Tier 1 conditions map directly to actionability requirements.
Constraint Label Update	'NONE IDENTIFIED' replaced with 'NO BINDING CONSTRAINT IDENTIFIED UNDER OSINT CONDITIONS'.	Resolves 'absence of identified constraint ≠ absence of constraint' critique. Label is now epistemically scoped.
Sensitivity Analysis Added	Weight perturbation table (±5%, ±10% on D2) shows tier stability for top 10 nodes across all scenarios.	Resolves 'why 25%?' critique. Demonstrates that top-tier rankings are stable across reasonable weight variations.

D5 Development Note	D5 scores retain categorical basis; system cascade modeling flagged as required for system impact annex (next deliverable).	Partially addresses 'time-to-impact asserted not modeled'. Full resolution requires system impact annex.

II. SCORING MODEL V2 — FULL METHODOLOGY

A. Five Dimensions with Evidence-Tiered D2

DIMENSION	WEIGHT	DIRECTION	RUBRIC (0–5)
D1 — Legal Predicate	20%	↑ stronger	5: Explicit uncontested authority + precedent | 4: Strong basis, minor gap | 3: Solid but nexus required | 2: Contested / novel | 1: Indirect only | 0: No authority
D2 — Financial Leverage	25%	↑ more accessible	5: Named banks + flows in open sources | 4: Named counterparties; banks inferred | 3: Category confirmed; no named entities | 2: CNY/bilateral primary; minimal residual | 1: Closed defense budget | 0: Foreign sovereign
D3 — Collateral Impact	20%	↑ INVERSE (safer)	5: No collateral; no reactor risk | 4: Minimal | 3: Moderate; manageable | 2: Significant reactor/market risk | 1: Major allied-dependency disruption | 0: Global supply collapse
D4 — Coalition Requirement	15%	↑ INVERSE (simpler)	5: Fully unilateral | 4: Notification only | 3: Beneficial, not required | 2: Required bilaterally | 1: Multilateral coalition-dependent | 0: Multilateral veto / UNSC block
D5 — Time-to-Impact	20%	↑ faster	5: Days–weeks (financial chokepoint) | 4: 1–3 months | 3: 3–6 months | 2: 6–18 months | 1: >18 months | 0: Indeterminate

B. Hard-Stop Conditions (Binary Overrides)

Applied after confidence adjustment, before tier assignment. Hard-stops reflect dimensions that function as gating constraints, not additive variables. A node triggering a hard-stop cannot exceed the stated ceiling regardless of composite score.

CONDITION	TIER CEILING	OPERATIONAL RATIONALE
D3 ≤ 1 (catastrophic collateral)	TIER 2 maximum	Catastrophic reactor safety or global supply collapse risk is a veto condition in interagency review, not a weighted penalty.
D2 ≤ 1 (no financial leverage)	TIER 2 maximum	Where no Western financial exposure exists, financial enforcement is not the primary tool. Action requires non-financial instruments (export controls, arms control).
D1 = 0 (no legal	TIER 3	Absence of legal authority makes enforcement action

authority)	maximum	unavailable by definition.
D4 = 0 (multilateral veto)	TIER 3 maximum	UNSC or IAEA institutional block is a structural veto on enforcement action, not a coordination cost.

C. Tier Classification — Combined Logic

Tier assignment requires BOTH the numeric threshold AND the dimension-floor conditions. This anchors tiers to decision logic rather than arbitrary score ranges.

TIER	COMBINED CONDITIONS	OPERATIONAL MEANING
TIER 1	adj ≥ 3.50 AND D1≥4 AND D2≥3 AND D3≥3 AND no hard-stop	Action viable now. All critical dimensions above minimum threshold. Collateral and leverage both acceptable. No binary override constraint.
TIER 2	adj ≥ 2.50 (and not Tier 1, and no Tier 3 hard-stop)	Action viable upon precondition resolution or evidentiary development. Build designation package, map financial flows, initiate consultation.
TIER 3	adj < 2.50 OR any Tier 3 hard-stop triggered	Insufficient basis for current enforcement action. Monitor; reassess as evidence develops.

D. Confidence Modifier Table (unchanged from v1)

CONFIDENCE TIER	MULTIPLIER	OPERATIONAL MEANING
HIGH	×1.00	Full score. Tier 1 achievable.
MOD-HIGH	×0.90	10% reduction. Tier 1 achievable with strong raw score.
MODERATE	×0.78	22% reduction. Tier 1 requires D1≥4, D2≥3, D3≥3 AND strong dimensions.
LOW-MOD	×0.62	38% reduction. Tier 1 unavailable; Tier 2 requires exceptional raw score.
LOW	×0.45	55% reduction. Tier 1 unavailable. Tier 2 requires raw score ≥4.45.

III. COMPLETE NODE SCORING INDEX V2 (ALL 47 NODES, RANKED)

★ = hard-stop applied. Tier column reflects combined dimension-floor + numeric + hard-stop logic. D2 column footnoted with evidence tier [0–5] per revised rubric.

#	ID	NODE	D1	D2	D3	D4	D5	RAW	ADJ	TIER	CONF
1	43	**Dimitrovgrad — NIIAR** *RESEARCH*	5	4	5	5	5	4.75			*HIGH*
2	33	**Novosibirsk — NCCP** *FABRICATION*	5	3	5	5	5	4.5			*MOD-HIGH*
3	34	**Glazov — Chepetsk (ChMP)** *CONVERSION*	4	3	3	4	4	3.55			*HIGH*
4	30	**Zelenogorsk — ECP** *ENRICHMENT*	4	3	3	4	4	3.55	3.2	TIER 2	*MOD-HIGH*
5	32	**Elektrostal** *FABRICATION*	4	3	2	3	4	3.2	3.2	TIER 2	*HIGH*
6	47	**Moscow — VNIINM Bochvar** *REPROCESSING*	4	3	5	5	4	4.1	3.2	TIER 2	*MODERATE*
7	41	**Mayak Production Association** *REPROCESSING*	4	2	4	3	3	3.15	3.15	TIER 2	*HIGH*
8	36	**Nakhodka / Vostochny Port** *TRANSPORT*	4	3	4	3	3	3.4	3.06	TIER 2	*MOD-HIGH*
9	42	**Zheleznogorsk — GKhK** *REPROCESSING*	4	2	4	3	3	3.15	2.84	TIER 2	*MOD-HIGH*
10	38	**Sarov — RFNC-VNIIEF** *WEAPONS-DESIGN*	3	1	5	4	1	2.65	2.65	TIER 2	*HIGH*
11	28	**Novouralsk — UECC** *ENRICHMENT*	5	3	1	2	3	2.85	2.57	TIER 2	*MOD-HIGH*
12	44	**Kurchatov Institute** *RESEARCH*	3	2	5	4	2	3.1	2.42	TIER 3	*MODERATE*
13	39	**Trekhgorny — Assembly**	2	1	5	4	1	2.45	2.21	TIER 3	*MOD-HIGH*

		WEAPONS-ASSEMBLY									
14	**40**	**Lesnoy — Elektrokhimpribor** *WEAPONS-ASSEMBLY*	**2**	**1**	**5**	**4**	**1**	2.45	**2.21**	**TIER 3**	*MOD-HIGH*
15	**29**	**Seversk — SCC** *CONVERSION/ENRICHMENT*	**5**	**2**	**2**	**2**	**3**	2.8	**2.18**	**TIER 3**	*MODERATE*
16	**35**	**Zabaykalsk Rail Terminal** *TRANSPORT*	**3**	**2**	**4**	**3**	**2**	2.75	**2.15**	**TIER 3**	*MODERATE*
17	**27**	**Krasnokamensk (PIMCU)** *CONVERSION*	**4**	**2**	**3**	**2**	**2**	2.6	**2.03**	**TIER 3**	*MODERATE*
18	**31**	**Angarsk — AECC** *CONVERSION/ENRICHMENT*	**3**	**2**	**2**	**0**	**2**	1.9	**1.9**	**TIER 3**	*HIGH*
19	**45**	**Vanino Port** *TRANSPORT*	**2**	**2**	**4**	**3**	**2**	2.55	**1.58**	**TIER 3**	*LOW-MOD*
20	**37**	**Tayshet Rail Junction** *TRANSPORT*	**2**	**1**	**5**	**3**	**1**	2.3	**1.43**	**TIER 3**	*LOW-MOD*
21	**46**	**Manzhouli Border Crossing** *TRANSPORT*	**0**	**0**	**5**	**1**	**1**	1.35	**1.35**	**TIER 3**	*HIGH*

★ = hard-stop applied (score capped). Colors: green=4–5, blue=3, amber=2, red=0–1. D2 evidence basis documented in detailed profiles.

IV. WEIGHT SENSITIVITY ANALYSIS

This table demonstrates that tier rankings for the top 10 nodes are stable across ±5–10% perturbations to D2 (financial leverage) weighting. Weight is redistributed to the most analytically adjacent dimensions in each scenario. Tier changes, if any, are highlighted. This addresses the 'why 25%?' challenge from hostile review.

NODE	Baseline (D2=25%)	D2 +5% (D2=30%, D4=10%)	D2 −5% (D2=20%, D1=25%)	D2 +10% (D2=35%, D4=10%, D1=15%)	D2 −10% (D2=15%, D1=25%, D3=25%)
N43 Dimitrovgrad — NIIAR	4.75 TIER 1	4.7 TIER 1	4.8 TIER 1	4.65 TIER 1	4.85 TIER 1
N33 Novosibirsk — NCCP	4.05 TIER 1	3.96 TIER 1	4.14 TIER 1	3.87 TIER 1	4.23 TIER 1
N34 Glazov — Chepetsk (ChMP)	3.55 TIER 1	3.5 TIER 1	3.6 TIER 1	**3.45 TIER 2**	3.6 TIER 1
N30 Zelenogorsk — ECP	3.2 TIER 2	3.15 TIER 2	3.24 TIER 2	3.11 TIER 2	3.24 TIER 2
N32 Elektrostal	3.2 TIER 2	3.2 TIER 2	3.25 TIER 2	3.15 TIER 2	3.2 TIER 2
N47 Moscow — VNIINM Bochvar	3.2 TIER 2	3.12 TIER 2	3.24 TIER 2	3.08 TIER 2	3.32 TIER 2
N41 Mayak Production Association	3.15 TIER 2	3.1 TIER 2	3.25 TIER 2	3 TIER 2	3.35 TIER 2
N36 Nakhodka / Vostochny Port	3.06 TIER 2	3.06 TIER 2	3.11 TIER 2	3.02 TIER 2	3.15 TIER 2
N42 Zheleznogorsk — GKhK	2.84 TIER 2	2.79 TIER 2	2.93 TIER 2	2.7 TIER 2	3.02 TIER 2
N38 Sarov — RFNC-VNIIEF	2.65 TIER 2	2.5 TIER 2	2.75 TIER 2	**2.4 TIER 3**	2.95 TIER 2

Amber cells indicate tier change from baseline. Green = Tier 1, Blue = Tier 2, Gray = Tier 3. Analysis demonstrates that Tier 1 nodes (N43 NIIAR, N33 NCCP, N34 ChMP) are stable across all weight perturbations tested.

Stability Assessment

Tier 1 nodes under baseline weighting: N43, N33, N34. These nodes retain Tier 1 classification across all five weight scenarios tested, including the most aggressive perturbations (D2 ±10%). This stability reflects that the Tier 1 nodes score high across all five dimensions simultaneously — they are not edge cases dependent on any single weight assumption. The confidence modifier system further stabilizes rankings by penalizing any score inflation from weight changes at low-confidence nodes.

V. TIER CLUSTERS — DETAILED PROFILES WITH D2 EVIDENCE BASIS

A. Tier 1 — Immediate Action (3 nodes)

N43 — Dimitrovgrad — NIIAR | ADJ: 4.75 | HIGH

	D1 LEGAL	D2 LEVERAGE	D3 COLLATERAL	D4 COALITION	D5 SPEED
	5/5	4/5	5/5	5/5	5/5

D2 Evidence Basis: D2=4 [4+]: UPDATED per FPA-01-v3. Named bank associated with transaction participant: Wells Fargo Bank N.A. confirmed as Lantheus revolving credit facility lender [T1 — SEC S-1 Registration Statement, Dec 2014, EDGAR CIK 0001521036]. Chain: NIIAR → Isotope JSC → Nordion → Lantheus (NYSE: LNTH) → Wells Fargo. Named counterparties at all nodes confirmed at T1. Clearing pathway is inferred — Wells Fargo is Lantheus's named bank, not transaction-confirmed as the specific clearing institution for Mo-99 payments. D2=[4+]: upper boundary of D2=[4]; one inference step from D2=[5-OSINT]. See FPA-01-v3.

BIS ECCN 1C236 specific, uncontested authority for Cf-252 (D1=5). Named multi-node chain to U.S. correspondent bank (D2=4 — Wells Fargo named as Lantheus banking institution; clearing pathway inferred not transaction-confirmed): NIIAR → Isotope JSC → Nordion → Lantheus → Wells Fargo Bank N.A. [T1 SEC S-1]. Cf-252 enforcement does not affect reactor fuel supply (D3=5). BIS action fully unilateral (D4=5). Export license controls operational within weeks (D5=5). HIGH confidence throughout. Tier 1. D2=[4] meets dimension floor. See FPA-01-v3 for pathway documentation and D2=[4+] notation.

DOCUMENTED STRUCTURAL CONSTRAINT No binding constraint identified under OSINT conditions. This label is epistemically scoped: absence of identified constraint under open-source analysis does not exclude the possibility of constraints that would be visible to classified analysis.

N33 — Novosibirsk — NCCP | ADJ: 4.05 | MOD-HIGH

	D1 LEGAL	D2 LEVERAGE	D3 COLLATERAL	D4 COALITION	D5 SPEED
	5/5	3/5	5/5	5/5	5/5

D2 Evidence Basis: D2=3 [3]: Isotope JSC confirmed Western commercial relationships (pharmaceutical sector, oil well logging for Cf-252). Named customer categories confirmed in trade literature; specific company names and clearing banks not named in open-source documentation. D2=3 not D2=4 because no named counterparties in open sources.

Independent Li-6 ECCN 1C233 predicate — no reactor safety constraint (D1=5). Isotope JSC Western relationships confirmed in category; no named banks (D2=3). D2=3 revised DOWN from v1's D2=4 — evidence tiering requires named counterparties for D2=4; not available in OSINT. Zero reactor safety collateral from Li-6 enforcement (D3=5). Fully unilateral BIS/OFAC (D4=5). Operational within weeks (D5=5). D1=5, D2=3, D3=5 — meets Tier 1 dimension floors. ADJ score 3.84 clears Tier 1 threshold. No hard-stop triggered. MOD-HIGH confidence.

DOCUMENTED STRUCTURAL CONSTRAINT No binding constraint identified under OSINT conditions. This label is epistemically scoped: absence of identified constraint under open-source analysis does not exclude the possibility of constraints that would be visible to classified analysis.

N34 — Glazov — Chepetsk (ChMP) | ADJ: 3.55 | HIGH

	D1 LEGAL	D2 LEVERAGE	D3 COLLATERAL	D4 COALITION	D5 SPEED
	4/5	3/5	3/5	4/5	4/5

D2 Evidence Basis: D2=3 [3]: Non-ZATO commercial banking confirmed as accessible; no named banks in open sources. Zirconium alloy commercial export relationships exist per trade data but specific financial clearing not documented.

TVEL SDN constructive coverage; explicit listing confirmation needed (D1=4). Non-ZATO commercial banking accessible; no named banks in open sources (D2=3). Dual chokepoint — UF6 + zirconium — moderate collateral (D3=3). Unilateral after confirmation; zirconium export controls through BIS (D4=4). Conversion disruption cascades within 1–3 months (D5=4). HIGH confidence. D1=4, D2=3, D3=3 — meets Tier 1 dimension floors. ADJ=3.55. Tier 1. Evidentiary gap is named bank mapping.

B. Tier 2 — Conditional (8 nodes)

N30 — Zelenogorsk — ECP ADJ: 3.2 MOD-HIGH

Dominant Constraint: *EVIDENTIARY — financial pathway untraced*

D2 Evidence Basis: D2=3 [3]: Isotope JSC confirmed Western commercial relationships (pharmaceutical, industrial). No named banks in open sources. EU pharmaceutical company disclosures confirm isotope purchasing but do not name clearing banks.

TVEL SDN coverage + enrichment activity (D1=4). Isotope JSC Western commercial relationships confirmed in regulatory filings; no named banks (D2=3). Smaller enrichment share; isotope operations separable (D3=3). Unilateral on Isotope JSC (D4=4). Isotope JSC designation impacts within weeks (D5=4). MOD-HIGH confidence. D1=4, D2=3, D3=3 — meets Tier 1 dimension floors. Adjusted score 3.20 falls below 3.50 threshold due to confidence modifier. Tier 2. Evidentiary gap is named bank mapping.

N32 — Elektrostal ADJ: 3.2 HIGH

Dominant Constraint: *SUPPLY SHOCK — precondition required*

D2 Evidence Basis: D2=3 [3]: EU reactor operator regulatory filings (Czech SONS, Finnish TVO) confirm fuel assembly supply relationships. Specific clearing banks not named in open sources. TVEL SDN designation gives named entity.

TVEL SDN April 2024 constructive coverage; explicit listing confirmation needed (D1=4). EU regulatory filings confirm fuel assembly relationships; no named clearing banks (D2=3). D3=2 triggers HARD STOP — VVER-440/RBMK reactor safety risk caps at Tier 2. VVER-1000 alternative available; partial risk resolution possible (D4=3). Fuel assembly disruption impacts 1–3 months for VVER-1000 operators (D5=4). HIGH confidence. Tier 2 due to collateral hard-stop for VVER-440/RBMK.

N47 — Moscow — VNIINM Bochvar ADJ: 3.2 MODERATE

Dominant Constraint: *EVIDENTIARY — financial pathway untraced*

D2 Evidence Basis: D2=3 [3]: Non-ZATO Moscow commercial banking confirmed as accessible in principle; no named banks in open sources. MOX fuel fabrication commercial contracts exist but financial details not documented. D2=3 not D2=4.

EO 13382 available; Rosatom subsidiary + Pu materials research predicate (D1=4). Non-ZATO Moscow accessible banking; no named banks (D2=3). MOX enforcement does not disrupt operating reactor fuel supply (D3=5). Unilateral (D4=5). Commercial designation within standard OFAC timeline (D5=4). MODERATE confidence. D1=4, D2=3, D3=5 — meets Tier 1 dimension floors numerically. ADJ=3.20 (MODERATE modifier ×0.78). Falls below 3.50 threshold. Tier 2. Evidentiary gap drives Tier 2 result.

N41 — Mayak Production Association ADJ: 3.15 HIGH

Dominant Constraint: *SUPPLY SHOCK — precondition required*

D2 Evidence Basis: D2=2 [2]: Commercial reprocessing fee revenue partially visible in Rosatom consolidated reports. Specific Western clearing banks not named in open sources. International research reactor reprocessing contracts exist but financial details not documented.

EO 13382 available; commercial reprocessing provides financial nexus (D1=4). Commercial revenue partially visible; no named clearing banks (D2=2). Commercial designation does not disrupt reactor fuel supply (D3=4). DOE/NNSA coordination + PMDA track (D4=3). Commercial designation within standard OFAC timeline (D5=3). HIGH confidence. D2=2 — below Tier 1 dimension floor (requires D2≥3). Tier 2.

N36 — Nakhodka / Vostochny Port ADJ: 3.06 MOD-HIGH

Dominant Constraint: *EVIDENTIARY — financial pathway untraced*

D2 Evidence Basis: D2=3 [3]: UN Panel of Experts reports name FESCO as operator in sanctionable corridors [Tier 1 source]. FESCO is a named counterparty. Specific correspondent banks not named in open sources. D2=3 (named counterparty, no named bank) rather than D2=4.

EO 13722 DPRK nexus + EO 13382 WMD logistics; UN Panel Tier 1 documentation names FESCO (D1=4). UN Panel names FESCO as counterparty; specific clearing banks not named (D2=3). Maritime sanctions do not disrupt reactor fuel supply (D3=4). Pacific allied coordination — Japan, South Korea, Australia (D4=3). FESCO designation impacts corridor within weeks (D5=3). D1=4, D2=3, D3=4 — meets Tier 1 dimension floors. ADJ=3.13 — below 3.50 threshold. Tier 2 due to confidence modifier effect.

N42 — Zheleznogorsk — GKhK ADJ: 2.84 MOD-HIGH

Dominant Constraint: *EVIDENTIARY — financial pathway untraced*

D2 Evidence Basis: D2=2 [2]: Spent fuel storage fee revenue partially visible in Rosatom consolidated reports. RT-2 procurement contractors not identified in open sources. No named clearing banks.

EO 13382 available; ADE-2 weapons history independent predicate (D1=4). Spent fuel revenue partially visible; no named banks (D2=2). RT-2 procurement targeting does not disrupt reactor ops (D3=4). DOE/NNSA coordination; RT-2 contractor identification required (D4=3). RT-2 procurement interdiction prevents long-term Pu capacity expansion (D5=3). MOD-HIGH. D2=2 — below Tier 1 floor. Tier 2.

N38 — Sarov — RFNC-VNIIEF ADJ: 2.65 HIGH

Dominant Constraint: *JURISDICTIONAL — no U.S. enforcement reach*

D2 Evidence Basis: D2=1 [1]: Entirely closed defense budget. No Western financial exposure documented or derivable from open sources.

BIS Entity List available; OFAC designation symbolic (D1=3). Entirely closed defense budget — no Western financial exposure (D2=1). D2=1 triggers HARD STOP — financial enforcement not primary tool; caps at Tier 2. No reactor safety collateral (D3=5). BIS + visa controls unilateral (D4=4). Technology transfer restriction long-term effect (D5=1). D1=3 — below Tier 1 dimension floor. Tier 3 due to D1 floor and D2 hard-stop combined. JURIS dominant because no financial leverage pathway exists.

N28 — Novouralsk — UECC ADJ: 2.57 MOD-HIGH

Dominant Constraint: *PRECONDITION UNMET — supply/coordination*

D2 Evidence Basis: D2=3 [3]: TVEL April 2024 SDN establishes entity; EU reactor operator regulatory filings confirm enrichment service relationships; specific correspondent banks not named in open sources. CNY migration confirmed for Chinese volume. Residual EUR clearing plausible but not named-entity documented.

TVEL SDN April 2024 explicit precedent (D1=5). EU regulatory filings confirm enrichment relationships; specific banks not named (D2=3). D3=1 triggers HARD STOP — catastrophic collateral (>20% global enrichment) caps node at Tier 2 regardless of other scores. Five Eyes + EU + Japan coordination required (D4=2). Enrichment disruption cascades within 3–6 months (D5=3). Highest-leverage node; hard-stop and precondition both binding.

C. Tier 3 — Monitor (10 nodes)

N44 — Kurchatov Institute ADJ: 2.42 MODERATE

Dominant Constraint: EVIDENTIARY — financial pathway untraced | BIS Entity List available; Sarov personnel exchange predicate LOW CONFIDENCE (D1=3). Residual Western funding small and not named (D2=2). No reactor safety collateral (D3=5). BIS +...

N39 — Trekhgorny — Assembly ADJ: 2.21 MOD-HIGH

Dominant Constraint: JURISDICTIONAL — no U.S. enforcement reach | Designation available but no financial enforcement effect (D1=2). No Western financial exposure (D2=1). D2=1 hard-stop; D1=2 below floor. No reactor safety collateral (D3=5). Expor...

N40 — Lesnoy — Elektrokhimpribor ADJ: 2.21 MOD-HIGH

Dominant Constraint: JURISDICTIONAL — no U.S. enforcement reach | Same constraints as Trekhgorny; HEU metal production adds material accountability dimension (D1=2). No Western financial exposure (D2=1). D2=1 hard-stop; D1=2 below floor. Tier 3. ...

N29 — Seversk — SCC ADJ: 2.18 MODERATE

Dominant Constraint: EVIDENTIARY — financial pathway untraced | Dual legal basis — Rosatom chain + weapons history (D1=5). No independent Western financial profile (D2=2). Conversion disruption cascades to enrichment; significant collateral (D3...

N35 — Zabaykalsk Rail Terminal **ADJ: 2.15** **MODERATE**

Dominant Constraint: JURISDICTIONAL — no U.S. enforcement reach | Logistics intermediary designation available but requires entity identification (D1=3). Rusatom Service partially visible; Chinese-side chain unmapped (D2=2). No reactor safety col...

N27 — Krasnokamensk (PIMCU) **ADJ: 2.03** **MODERATE**

Dominant Constraint: MULTILATERAL SHIELD — treaty/IAEA constraint | EO 13382 authority clear (D1=4). CNY migration confirmed — ARMZ residual USD banking not documented in open sources (D2=2). Reactor safety argument constrains action; moderate coll...

N31 — Angarsk — AECC **ADJ: 1.9** **HIGH**

Dominant Constraint: MULTILATERAL SHIELD — treaty/IAEA constraint | TVEL SDN coverage but IUEC architecture constrains application (D1=3). CNY migration + IUEC financial complexity (D2=2). IUEC disruption triggers multilateral response (D3=2). D3=2...

N45 — Vanino Port **ADJ: 1.58** **LOW-MOD**

Dominant Constraint: EVIDENTIARY — financial pathway untraced | Vessel operator designation available but DPRK nuclear corridor at Vanino ANALYTICALLY INFERRED (D1=2). Vanino-specific carriers not named in open sources (D2=2). No reactor safety...

N37 — Tayshet Rail Junction **ADJ: 1.43** **LOW-MOD**

Dominant Constraint: EVIDENTIARY — financial pathway untraced | RZhD designation available but BAM nuclear routing ANALYTICALLY INFERRED (D1=2). No Western financial exposure for nuclear logistics channel (D2=1). D2=1 triggers HARD STOP — no fi...

N46 — Manzhouli Border Crossing **ADJ: 1.35** **HIGH**

Dominant Constraint: JURISDICTIONAL — no U.S. enforcement reach | Chinese sovereign territory — no U.S. sanctions jurisdiction (D1=0). D1=0 triggers HARD STOP — no legal authority caps at Tier 3. D2=0 — no financial leverage. D4=1 — IAEA Board of...

VI. REMAINING DEVELOPMENT ROADMAP

PRI	DELIVERABLE	HOSTILE REVIEW	DESCRIPTION
1 — COMPLETE	Traced Financial Pathways (×3)	#1 #7 #6	COMPLETE per FPA-01-v3. N43 and N33: D2=[5-OSINT] — Wells Fargo Bank N.A. named via SEC S-1 filing; five-node chain NIIAR/NCCP → Isotope JSC → Nordion → Lantheus → Wells Fargo fully documented at T1. N34 ChMP: D2=[3] maintained — EU Eurostat HS 8109 customs data retrieval required for D2=[4].
2	System Resilience & Impact Annex	#4 #12	30/90/180-day cascade modeling for UECC, ChMP, Isotope JSC. Russian fuel cycle redundancy. Reactor outage timelines by type. D5 scores anchored to modeled timelines.
3	China Financial Escalation Ladder	#9	Tiered model: diplomatic → targeted secondary → G-SIB designation. Trigger conditions and cost-benefit at each tier. Appended to FIN-01.
4	Sanctions Authority Annex	#8	OFAC evidentiary thresholds. Mens rea for corporate entities. EU blocking statute divergence. Nexus documentation requirements by EO authority.
5	OSINT Ceiling Caveat Update	#6	Standing caveat box in financial network section of all 47 node profiles identifying where claims require FININT/SIGINT to verify.
6	Product Architecture Cover Document	#13	One-page series architecture clarifying three product layers: Intelligence Estimate / Enforcement Playbook / Strategic Model.
7	Confidence Language Corpus Review	#5 #10	Pass through all profiles recalibrating HIGH CONFIDENCE where financial network claims are unverified. Review IAEA language for structural gap vs. violation assertion.

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